Helix OTA — Feature Inventory — Status Summary

Revision: 1 **Last modified:** 2026-06-18T12:00:00Z **Companion of:** `Status.md` (Section 11.4.56 two-audience parity).

Page 1 — For the operator (plain language)

Helix OTA is an enterprise over-the-air update system. Here is what exists and what state it is in.

What works today:

- The **control-plane server** (Go/Gin) is fully built and tested — all APIs for managing devices, releases, deployments, rollouts, recalls, and audit logs have both unit tests and end-to-end tests running in containers.
- **Six reusable Go libraries** (protocol types, artifact validation, rollout engine, telemetry, HTTP/3) are built and tested with unit + stress + chaos tests.
- **Container infrastructure** is in place via the `vasic-digital/containers` submodule — tests run in podman pods.
- **The emulator ladder proves the A/B update core on this Mac without needing the real hardware:**
  - The container round-trip (control plane talking to a virtual device) works.
  - The emulated device boots to a live Linux userspace — full boot transcript captured as proof.
  - **The A/B slot switch is proven** — a real U-Boot bootloader switches between system slot A and slot B on demand, with 3/3 identical passes.
  - **Automatic rollback on a corrupt slot is proven** — when we mark a slot bad, the bootloader detects it and falls back to the good slot automatically.
- **Security tests**, **e2e tests**, and **pre-build gates** all pass.
- **Governance** (Issues, Fixed, CONTINUATION, README, Status docs) is maintained and in sync.

What is still pending (NOT done yet — honestly):

- **RAUC dm-verity tamper protection** — the test script is written but not yet run. It needs a signed update bundle and some slot-scheme alignment.
- **ApplyPort (the actual Android OTA apply)** — the design is documented but not built yet. This will bridge the server to the Android update engine.
- **Android OTA agent modules** — the two Kotlin modules exist as scaffolds but have not been tested against a real Android target.
- **Full Android OTA test (Tier-2 Cuttlefish)** — cannot run on this Mac (needs Linux with KVM). Ready for the operator’s Linux machine.
- **Real hardware testing (Tier-3 RK3588 board)** — no board on the bench yet.

- **Stress/chaos coverage** — present for some submodules but not yet for the server endpoints.
- **Build resource statistics tracking** — mandated but not yet built.
- **Workable-items SQLite database** — mandated but not yet built.
- **CodeGraph MCP integration** — mandated but not yet installed.

Bottom line: The server, Go libraries, and the A/B update core (slot switch + auto-rollback) are proven with captured evidence. The Android-specific apply layer, dm-verity integrity, and hardware-tier tests remain as the open items before a release tag.

Page 2 — For software engineers

Feature inventory summary (all 89 items from Status.md):

Status	Count	Key Items
PASS	40	All server handlers (F01-F34), emulator Tier-0/ Tier-1 (F44-F49), e2e tests (F57-F67), build gates (F69-F71), scripts (F74-F76)
VERIFIED	14	Go submodules (F35-F41), containers submodule (F68), .gitignore (F73), governance doc set (F77-F85)
PROVEN	3	PWU-AB-1 base+boot (F50), PWU-AB-1 slot switch (F51), PWU-AB-3 auto-rollback (F52)
PENDING_FORENSICS	1	PWU-AB-2 RAUC dm-verity (F53)
DESIGN	3	ota-android-agent (F42), ota-update-engine-bridge (F43), PWU-AB-4 ApplyPort (F54)
OPERATOR-BLOCKED	2	Tier-2 Cuttlefish (F55 — needs Linux+KVM), Tier-3 HW (F56 — needs board)
PARTIAL	2	Stress+chaos coverage (F86 — 2/12 submodules), Docs Chain (F89 — engine built, not submoduled)
NOT_STARTED	3	Build-resource-stats (F72), workable-items

Status	Count	Key Items
		DB (F87), CodeGraph (F88)

Proven A/B core (captured evidence): - PWU-AB-1 slot switch — `docs/qa/20260611T094958Z-ab-slot-switch/` (3/3 deterministic PASS, real U-Boot 2024.01 on QEMU virt + HVF) - **PWU-AB-3 auto-rollback** — `docs/qa/20260611T095918Z-ab-rollback/` (bad slot -> bootcount exceeded -> altbootcmd swap -> known-good slot; CONTROL proves rollback only fires on bad slot)

Pending (honest per Section 11.4.6): - PWU-AB-2 RAUC dm-verity: script authored (`ab_rauc_verity.sh`), gated on signed `.raucb` bundle + slot-scheme reconciliation - PWU-AB-4 ApplyPort: design doc exists, code not yet built - Tier-2: `tier2_cuttlefish_ab.sh` authored, SKIPs on this host (no `/dev/kvm`) - Tier-3: No physical board

No release tag — Section 11.4.40 requires full ladder GREEN; Tier-2 + Tier-3 remain blocked.

Video Recording Gaps

Priority	Gap	Depends On
High	Tier-2 Cuttlefish Android OTA apply screen recording	Linux+KVM host
High	Tier-3 RK3588 HDMI capture of on-device OTA	Physical board
Medium	Tier-1 AVD + HVF screen recording	Existing qa evidence may be partial
Low	Tier-0 container round-trip	Console logs suffice
Low	A/B slot switch + rollback	Console transcripts are definitive
Future	Web UI screen recording	Web UI not yet built
N/A	Audio routing	No audio subsystem in scope

Section-refs

Section 11.4.45 (Status doc), Section 11.4.56 (two-audience summary), Section 11.4.5 (captured-evidence table), Section 11.4.6 (no-guessing), Section 11.4.44 (revision header), Section 11.4.65 (universal export).